

ERICK CHICA

Doctoral Candidate in Health Technology · Marie Skłodowska-Curie Fellow

📍 Copenhagen, Denmark • ✉ eccca@dtu.dk • 🌐 erickchica.com • [in LinkedIn](#) • 🎓 [Google Scholar](#)

»»» EDUCATION

TECHNICAL UNIVERSITY OF DENMARK - DTU

Dec 2025 - Present

Doctor of Philosophy (PhD) in Health Technology

📍 Copenhagen, Denmark

- **Research Area:** Micro/Nanofabrication, Ingestible Devices, Drug Release, Bioelectronics.
- **Funding:** Marie Skłodowska-Curie Actions – INTELLI-INGEST Doctoral Network.

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY - KAIST

Feb 2022 - Aug 2025

Master of Science (MS) in Electrical Engineering

📍 Daejeon, South Korea

- **Research Area:** Bioelectronic Systems, Flexible Electronics, Organ-on-a-Chip and Organoids.
- **Thesis:** Development of a Three Dimensional Multiwell Millifluidic Platform Integrated with Stretchable Microelectrode Arrays for Organoid Evaluation in Real Time.
- **Funding:** KAIST Scholarship for International Students.

KOREA UNIVERSITY

Mar 2018 - Feb 2022

Bachelor of Engineering (BEng) in Electrical Engineering

📍 Seoul, South Korea

- **Graduation Project:** Design of a Fault Detecting Device for Vibrating Machines.
- **Funding:** Global Korea Scholarship (GKS).

»»» RESEARCH EXPERIENCE

DTU - DEPARTMENT OF HEALTH TECHNOLOGY

Dec 2025 - Present

Doctoral Candidate

📍 Copenhagen, Denmark

Summary: Developing microfabricated ingestible devices for controlled drug release within the INTELLI-INGEST MSCA Doctoral Network.

- Develop microfabrication processes for ingestible capsule prototypes, including clean-room photolithography, thin-film deposition, and device integration workflows.
- Design and iteration of capsule architectures for controlled drug release, incorporating insights from consortium-wide collaboration across partner institutions.

KAIST - SCHOOL OF ELECTRICAL ENGINEERING

Feb 2022 - Aug 2025

Graduate Researcher

📍 Daejeon, South Korea

Summary: Developed a millifluidic platform integrating three stretchable microelectrode arrays for parallel electrophysiology of brain organoids under uniform, standardized perfusion.

- Designed and fabricated integrated PDMS/SU-8 microfluidic and millifluidic chips with embedded microelectrode arrays, optimizing channel geometry for uniform media perfusion across multiple organoid chambers.
- Recorded electrophysiological activity simultaneously from three human midbrain organoids under continuous perfusion, and validated the flow profiles computationally and experimentally.
- Investigated and prototyped flexible biocompatible materials and fabrication techniques for next-generation wearable sensors and organ-on-chip applications.

»»» PUBLICATIONS AND MANUSCRIPTS

1. **Chica, E.[†]**, Kwon, Y.[†], Jang, H., Kim, Y., Lee, Y., Kim, K., Gogos, J. A., Lee, M. & Lee, H. J.^{*} (2026). *Millifluidic triMEA organoid platform with stretchable microelectrode arrays for parallel organoid electrophysiology under standardized perfusion*. Lab on a Chip. doi.org/10.1039/d6lc00391e
2. Maric, T.^{*}, Soufi, G.^{*}, Li, Z., **Chica, E.**, Jinow, A. A. A., Al-Roubai, N., Ahmed, N., Göksel, Y. & Boisen, A. (2026). *Near-Infrared Light-Driven Zn/Au Janus Micromotors for Multiplex SERS Detection of Anticancer Drugs*. Advanced Functional Materials, 36(58), e75935. doi.org/10.1002/adfm.75935
3. Kim, Y.[†], **Chica-Carrillo, E.[†]** & Lee, H. J.^{*} (2024). *Microfabricated sensors for non-invasive, real-time monitoring of organoids*. Micro and Nano Syst Lett, 12(1), 26. doi.org/10.1186/s40486-024-00216-y

»»» CONFERENCE PRESENTATIONS

1. **Chica, E.** *Gold nanoparticle-functionalized Zn/Au and Mg/Au Janus micromotors for multiplex SERS detection of anticancer drugs*. Oral presentation at the Manipulation, Automation and Robotics at Small Scales (MARSS) 2026 Conference, Hangzhou, China, 30 July 2026.
2. **Chica, E.**, Kwon, Y., Jang, H., Kim, Y., Lee, Y., Lee, M., Lee, H.^{*} *Millifluidic triMEA organoid platform with stretchable microelectrode arrays for multi organoid evaluation*. Poster presentation at the Society of Micro and Nano Systems 2025 MNS Fall Conference, Yeosu, South Korea, 19 November 2025.

»»» SERVICE & LEADERSHIP

INTELLI-INGEST MSCA DOCTORAL NETWORK

Mar 2026 - Mar 2027

Doctoral Candidate Board Representative

📍 Copenhagen, Denmark

- Elected by peers to represent 13 doctoral candidates (DCs) across the INTELLI-INGEST consortium, serving as primary liaison between DCs and the network's supervisory board.
- Facilitate regular DC board meetings, consolidate collective feedback, and advocate on matters of training, welfare, and programme governance across partner institutions.

»»» TEACHING EXPERIENCE

KAIST - SCHOOL OF DIGITAL HUMANITIES AND SOCIAL SCIENCES

Feb 2024 - Aug 2025

Teaching Assistant: *Scientific Writing*

📍 Daejeon, South Korea

- Reviewed and graded 70+ student manuscripts across three semesters, providing detailed written feedback on structure, argumentation, and academic style.
- Worked with instructor to develop rubrics, plan course activities, and maintain online course platform.

KAIST - ENGLISH AS A FOREIGN LANGUAGE PROGRAM

Aug 2024 - Jun 2025

Academic Tutor

📍 Daejeon, South Korea

- Provided one-to-one tutoring sessions to 30+ undergraduate and graduate students to strengthen academic writing and speaking skills.

KAIST - SCHOOL OF ELECTRICAL ENGINEERING

Aug 2024 - Jun 2025

Teaching Assistant: *Nanobioelectronics*

📍 Daejeon, South Korea

- Evaluated presentations, assignments, and examinations for a class of 40+ students.
- Prepared and distributed supplementary materials on sensors, microfluidics, and electrochemistry.